

NETWORK TESTING TOOLS THEORY

1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.



Command Used:

ping www.youtube.com

Result:

```
Pinging www.youtube.com [2001:4860:4829:400::] with 32 bytes of data:
Reply from 2001:4860:4829:400::: time=31ms
Reply from 2001:4860:4829:400::: time=52ms
Reply from 2001:4860:4829:400::: time=75ms
Reply from 2001:4860:4829:400::: time=62ms

Ping statistics for 2001:4860:4829:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 31ms, Maximum = 75ms, Average = 55ms
```

2. Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.
Hint: Each hop represents a network device your data passes through to reach the destination.



Command Used:

tracert www.flipkart.com

```
Tracing route to flipkart.com [163.53.76.86]
over a maximum of 30 hops:

  1     1 ms      1 ms      1 ms    192.168.1.1
  2    40 ms     38 ms     37 ms    10.240.170.222
  3      *        *         *        Request timed out.
  4    41 ms     22 ms     36 ms    125.21.143.117
  5    44 ms     39 ms     38 ms    116.119.46.40
  6    82 ms     78 ms     94 ms    aes-static-082.27.22.125.airtel.in [125.22.27.82]
  7      *        *         *        Request timed out.
  8      *        *         *        Request timed out.
  9      *        *         *        Request timed out.
 10     *        *         *        Request timed out.
 11    76 ms     66 ms     60 ms    163.53.76.86

Trace complete.
```

3. Use the nslookup tool to find the IP address of **www.zomato.com** and record the result.



Command Used :

nslookup www.zomato.com

```
Server: UnKnown
Address: 2401:4900:50:9::7a3

Non-authoritative answer:
Name: www.zomato.com.domain.name
Addresses: 185.38.109.200
           185.38.109.201
           185.38.109.202
           185.38.109.203
           185.38.109.204
           185.38.109.205
           185.38.109.206
           185.38.109.207
           185.38.109.208
           185.38.109.209
```

4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.



Command Used:

ping 192.168.1.105

Pinging 192.168.1.105 with 32 bytes of data:

Reply from 192.168.1.105: bytes=32 time=4ms TTL=64

Reply from 192.168.1.105: bytes=32 time=5ms TTL=64

Reply from 192.168.1.105: bytes=32 time=3ms TTL=64

Reply from 192.168.1.105: bytes=32 time=4ms TTL=64

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Average response time: 4ms

If the device is offline:

Request timed out.

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.



Network Diagnostic Tools Report

```
Pinging www.youtube.com [2001:4860:4829:400::] with 32 bytes of data:
Reply from 2001:4860:4829:400::: time=31ms
Reply from 2001:4860:4829:400::: time=52ms
Reply from 2001:4860:4829:400::: time=75ms
Reply from 2001:4860:4829:400::: time=62ms

Ping statistics for 2001:4860:4829:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 31ms, Maximum = 75ms, Average = 55ms
```

Ping is useful for checking connectivity problems, such as a website not loading. For example, I used **ping** www.youtube.com and checked the response time and packet loss.

```
Tracing route to flipkart.com [163.53.76.86]
over a maximum of 30 hops:
  0  1 ms  1 ms  1 ms  192.168.1.1
  1  40 ms  38 ms  37 ms  10.240.170.222
  2  *      *      *      Request timed out.
  3  41 ms  22 ms  36 ms  125.21.143.117
  4  44 ms  39 ms  38 ms  116.119.46.40
  5  82 ms  78 ms  94 ms  aes-static-082.27.22.125.airtel.in [125.22.27.82]
  6  *      *      *      Request timed out.
  7  *      *      *      Request timed out.
  8  *      *      *      Request timed out.
  9  *      *      *      Request timed out.
 10 *      *      *      Request timed out.
 11 76 ms  66 ms  60 ms  163.53.76.86

Trace complete.
```

Traceroute (tracert) is useful for finding the location of delays in a slow connection. I used **tracert** www.flipkart.com to view the different network hops between my computer and the website.

```
Server: Unknown
Address: 2401:4900:50:9::7a3

Non-authoritative answer:
Name: www.zomato.com.domain.name
Addresses: 185.38.109.200
           185.38.109.201
           185.38.109.202
           185.38.109.203
           185.38.109.204
           185.38.109.205
           185.38.109.206
           185.38.109.207
           185.38.109.208
           185.38.109.209
```

Nslookup is useful for identifying DNS-related issues. I used **nslookup** www.zomato.com to find the IP address connected with the domain name.

By using these tools together, we can identify whether the problem is related to connectivity, network routing, or DNS resolution.